

# Kumar Abhishek

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## PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in **Machine Learning**, **Deep Learning**, **Generative AI**, and **Data Analytics**. Experienced in building production-ready AI systems using **PyTorch**, **FastAPI**, and modern AI frameworks, with projects spanning predictive modeling, statistical analysis, deep learning, LLM applications, and intelligent automation. Passionate about developing AI-driven solutions that transform data into actionable insights and solve real-world business problems.

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Indian Institute of Information Technology, Una</b><br><i>B.Tech, Electronics &amp; Communication Engineering</i> | Jul 2023 – Jul 2027<br>CGPA: 7.74/10 |
| <b>Delhi Model Public School, Delhi</b><br><i>Class XII (CBSE)</i>                                                   | Jun 2022<br>Score: 85%               |

## PROJECT EXPERIENCE

- InterviewBot – AI-Powered Interview Preparation Platform** Aug 2024 – Oct 2024  
*FastAPI | LangChain | LangGraph | Google Gemini | ChromaDB | MongoDB |* GitHub | Live
  - Built a production-ready AI interview platform comprising multiple FastAPI microservices with **LangGraph** workflows supporting RAG chat, interview setup, contextual assistance, and personalized interview evaluation.
  - Implemented **RAG pipeline** with ChromaDB vector store and **all-MiniLM-L6-v2** Sentence Transformers for semantic search over website-indexed content; built a resume parser (PDF/DOCX via LangChain) that extracts skills and generates personalized interview questions.
  - Delivered MCQ test engine and mock interview module with **JWT-secured FastAPI backend**, async MongoDB (Motor) storage, voice input via React Speech Recognition, and auto-generated **PDF performance reports** using ReportLab.
- Deep Learning for ECG Signal Analysis and R-Peak Detection** Sep 2025 – Dec 2025  
*PyTorch | Deep Learning | NumPy | SciPy | Scikit-learn | PyWavelets* GitHub
  - Implemented and benchmarked **6 R-peak detection approaches** on the **MIT-BIH Arrhythmia Database** – spanning classical signal processing (Pan-Tompkins, Wavelet Transform) and deep learning models (1D CNN, LSTM, IncRes-UNet/RPNet, Distance Transform Learning).
  - Achieved **99.5% sensitivity** with the Pan-Tompkins algorithm and **99.7% sensitivity / 99.6% F1-score** with the Inception-Residual UNet (RPNet), employing distance transform regression for improved gradient flow and robustness to noisy signals.
  - Designed and trained an **Inception-Residual U-Net (RPNet)** in PyTorch using a distance transform regression objective, enabling robust feature learning and accurate R-peak localization in noisy ECG signals.
- DataForge – AI-Powered Data Science & Business Analytics Platform** Mar 2026 – Jun 2026  
*Python | Streamlit | Pandas | Plotly | Scikit-learn | SHAP | LangChain | Google Gemini* GitHub | Live
  - Built an end-to-end **analytics platform** automating EDA, statistical hypothesis testing, feature engineering, AutoML, and business intelligence reporting workflows.
  - Developed an AI-powered **analytics assistant** (LangChain + Google Gemini) enabling natural language queries, automated insight generation, and executive-ready business recommendations.
  - Implemented automated **hypothesis testing** (t-test, ANOVA, Chi-Square) and interactive **KPI dashboards** with Plotly for data-driven decision-making and trend analysis.
  - Trained and compared multiple ML models with **AutoML**, using Explainable AI (**SHAP**) to surface the key business drivers behind each prediction.
  - Automated executive-ready **PDF report** generation by consolidating statistical findings, visualizations, model performance, and AI-generated insights into a single downloadable report.

## TECHNICAL SKILLS

- Programming:** Python, SQL.
- AI / ML / GenAI:** Machine Learning, Deep Learning, CNN, LSTMs, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, LangChain, LangGraph, LangSmith, MCP (Model Context Protocol), Transformers, Vector Databases, Agent-to-Agent (A2A) Protocol .
- Data Analysis & BI:** Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Data Cleaning, KPI Development, Dashboard Design, Power BI, Excel (Advanced).
- Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Transformers, FastAPI, Streamlit, SciPy, ReportLab.
- Backend & Databases:** FastAPI, FastMCP, MongoDB, PostgreSQL, ChromaDB
- Tools & Platforms:** GitHub, Jupyter Notebook, Vercel, AWS (EC2, S3, IAM – Basic), Hugging Face.
- Coursework:** Machine Learning, Probability & Statistics, Linear Algebra.

## POSITION OF RESPONSIBILITY

- Active Member**, AAVESH Technical Club, IIIT Una 2023 – Present
- Active Member**, EUNOIA Literature Club, IIIT Una 2023 – Present